Exp. No: 10

Verification of Faraday's Law of Electromagnetic Induction – Determination and Plotting of Induced EMF (Transformer EMF and Motional EMF)

Aim:

To verify Faraday's law of electromagnetic induction by determining and plotting the emf induced in a multi-turn loop due to (i) a time-varying magnetic flux density (transformer emf), (ii) conductor motion in a magnetic field (motional emf) and (iii) loop rotation (generator emf), and to confirm the polarity of the induced emf using Lenz's law.

Objectives:

1. To compute the magnetic flux linkage of a multi-turn loop placed in a sinusoidally varying magnetic flux density.
2. To determine the induced emf from $\text{emf} = -N\, d\Phi/dt$ both numerically and analytically and compare the two results.
3. To determine the motional emf of a conductor moving in a magnetic field and the emf generated by a rotating loop.
4. To investigate the effect of frequency, number of turns, velocity and angle on the induced emf.

Test Case (i) – Transformer EMF Induced by a Time-Varying Flux Density

MATLAB Program:

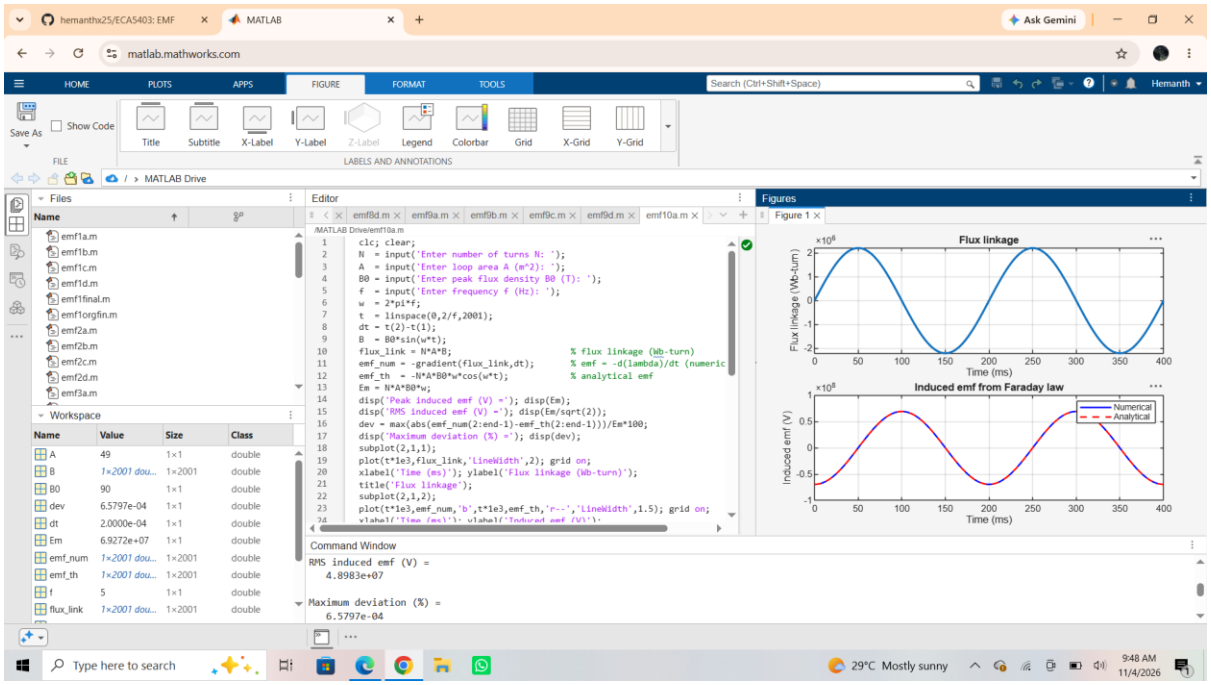
Test Case (i) – Transformer EMF Induced by a Time-Varying Flux Density

```
clc; clear;
N  = input('Enter number of turns N: ');
A  = input('Enter loop area A (m^2): ');
B0 = input('Enter peak flux density B0 (T): ');
f  = input('Enter frequency f (Hz): ');
w  = 2*pi*f;
t  = linspace(0,2/f,2001);
dt = t(2)-t(1);
B  = B0*sin(w*t);
flux_link = N*A*B;                  % flux linkage (Wb-turn)
emf_num = -gradient(flux_link,dt);      % emf = -d(lambda)/dt (numerical)
```

```
emf_th  = -N*A*B0*w*cos(w*t);          % analytical emf
Em = N*A*B0*w;
disp('Peak induced emf (V) ='); disp(Em);
disp('RMS induced emf (V) ='); disp(Em/sqrt(2));
dev = max(abs(emf_num(2:end-1)-emf_th(2:end-1)))/Em*100;
disp('Maximum deviation (%) ='); disp(dev);
subplot(2,1,1);
plot(t*1e3,flux_link,'LineWidth',2); grid on;
xlabel('Time (ms)'); ylabel('Flux linkage (Wb-turn)');
title('Flux linkage');
subplot(2,1,2);
plot(t*1e3,emf_num,'b',t*1e3,emf_th,'r--','LineWidth',1.5); grid on;
xlabel('Time (ms)'); ylabel('Induced emf (V)');
legend('Numerical','Analytical');
title('Induced emf from Faraday law');
```

Sample Output: For N = 100, A = 0.01 m², B0 = 0.5 T and f = 50 Hz, the peak emf = 157.08 V, the rms emf = 111.07 V and the maximum deviation is below 0.001 %.

Output:

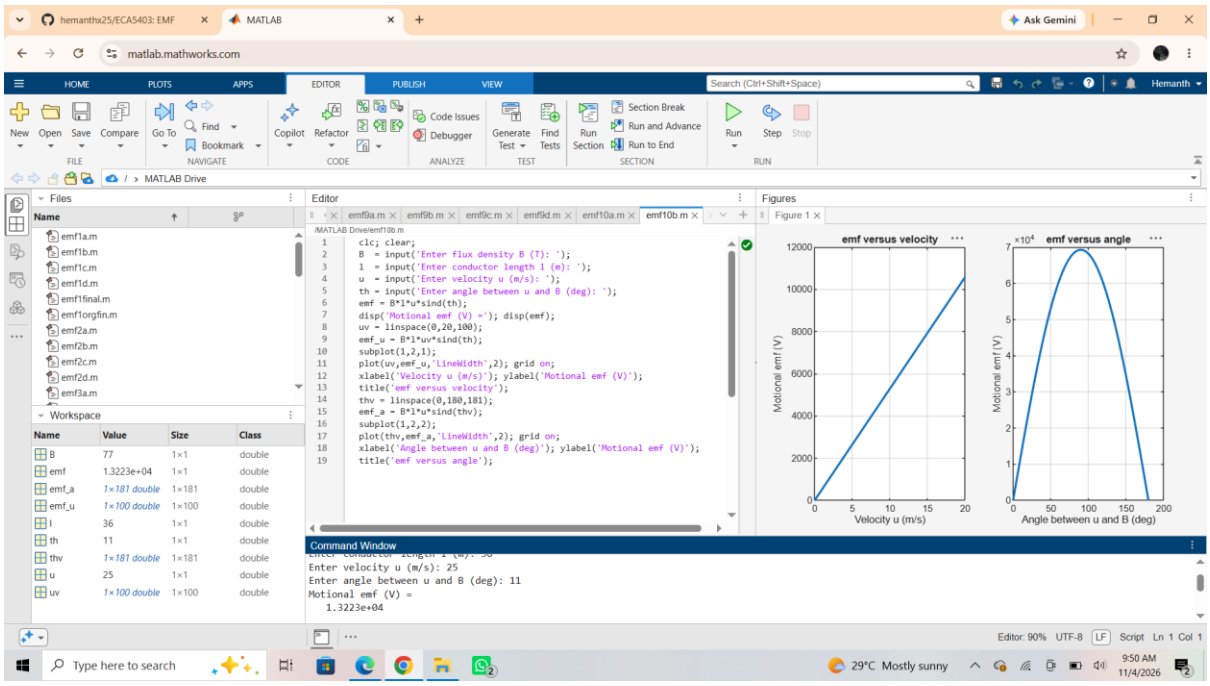


Test Case (ii) – Motional EMF of a Conductor Moving in a Magnetic Field

MATLAB Program:

```
clc; clear;
B  = input('Enter flux density B (T): ');
l  = input('Enter conductor length l (m): ');
u  = input('Enter velocity u (m/s): ');
th = input('Enter angle between u and B (deg): ');
emf = B*l*u*sind(th);
disp('Motional emf (V) ='); disp(emf);
uv = linspace(0,20,100);
emf_u = B*l*uv*sind(th);
subplot(1,2,1);
plot(uv,emf_u,'LineWidth',2); grid on;
xlabel('Velocity u (m/s)'); ylabel('Motional emf (V)');
title('emf versus velocity');
thv = linspace(0,180,181);
emf_a = B*l*u*sind(thv);
subplot(1,2,2);
plot(thv,emf_a,'LineWidth',2); grid on;
xlabel('Angle between u and B (deg)'); ylabel('Motional emf (V)');
title('emf versus angle');
```

Output:

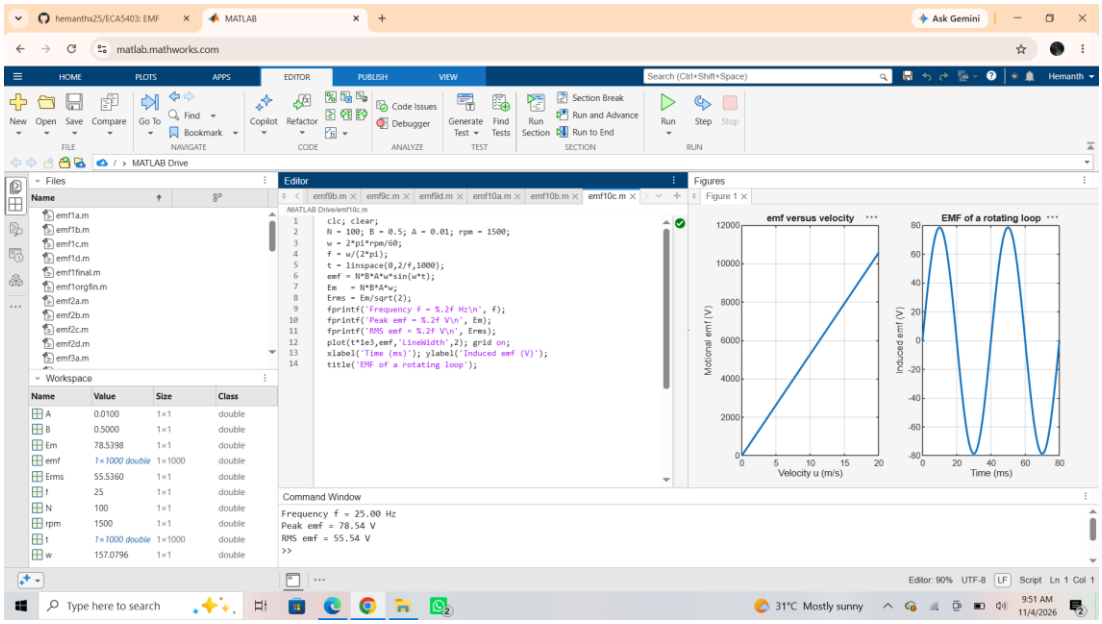


Test Case (iii) – EMF of a Rotating Loop (Generator Action)

```
clc; clear;
N = 100; B = 0.5; A = 0.01; rpm = 1500;
w = 2*pi*rpm/60;
f = w/(2*pi);
```

```
t = linspace(0,2/f,1000);
emf = N*B*A*w*sin(w*t);
Em   = N*B*A*w;
Erms = Em/sqrt(2);
fprintf('Frequency f = %.2f Hz\n', f);
fprintf('Peak emf = %.2f V\n', Em);
fprintf('RMS emf = %.2f V\n', Erms);
plot(t*1e3,emf,'LineWidth',2); grid on;
xlabel('Time (ms)'); ylabel('Induced emf (V)');
title('EMF of a rotating loop');
```

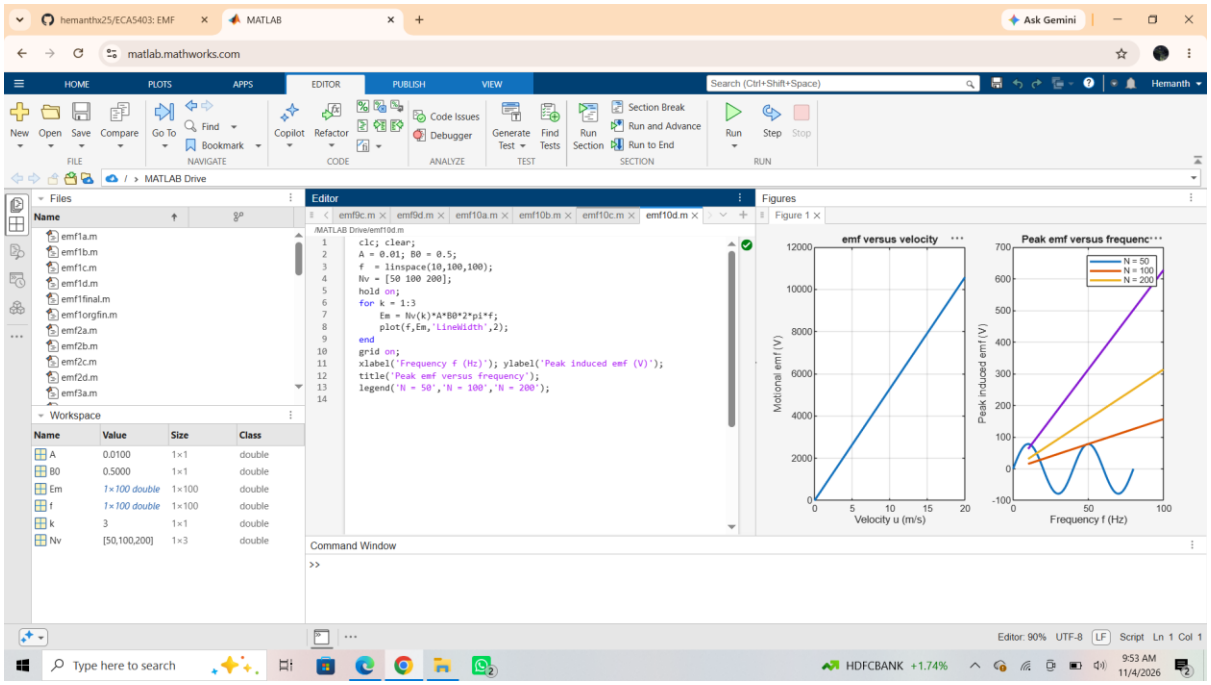
Output:



Test Case (iv) – Effect of Frequency and Number of Turns on the Peak EMF

```
clc; clear;
A = 0.01; B0 = 0.5;
f  = linspace(10,100,100);
Nv = [50 100 200];
hold on;
for k = 1:3
    Em = Nv(k)*A*B0*2*pi*f;
    plot(f,Em,'LineWidth',2);
end
grid on;
xlabel('Frequency f (Hz)'); ylabel('Peak induced emf (V)');
title('Peak emf versus frequency');
legend('N = 50','N = 100','N = 200');
```

Output:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify induction mechanism and parameters	Identify induction mechanism and parameters	Correctly identifies flux change/turns for transformer EMF and B, length, velocity/orientation for motional EMF	Identifies most parameters correctly	Some parameters or directions are unclear	Uses an incorrect physical model.	25	2.5	
2	Calculate induced EMF in MATLAB	Calculate induced EMF in MATLAB	Implements the applicable Faraday and motional EMF equations correctly with units/sign convention.	Calculations are mostly correct with minor errors.	Some calculations require assistance.	EMF calculations are incorrect.	25	2.5	
3	Plot and verify EMF behaviour	Plot and verify EMF behaviour	Plots expected variation and verifies results against theoretical relations.	Plots and verification are mostly correct	Plot or validation is incomplete.	No valid plot or verification.	30	3	
4	Interpret and document results	Interpret and document results	Distinguishes transformer and motional EMF and provides clear	Explanation is correct with	Explanation is limited.	Results are not explained or documented.	20	2	

			tables/plots/conc lusion	minor omissio ns.					
Total							100	10	